HARSHVARDHAN

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PROFESSIONAL SUMMARY

Software Engineer specializing in AI, with hands-on experience in LLMs, RAG systems, vector databases, and full-stack development using Python, Flask, and React. Skilled in building and deploying on-device LLM inference with Ollama, developing scalable AI-driven applications, and integrating deep learning models into production-ready systems.

TECHNICAL SKILLS

- Programming Languages: Python, JavaScript, C++
- Frontend: React, HTML, Tailwind CSS, CSS, JavaScript
- Backend Support: Flask, PostgreSQL, MongoDB(basic)
- Framework & Libraries: Pytorch, Flask, Langchain, Qdrant, LangGraph
- Tools & Technologies: GenAI, Ollama, Huggingface, Docker, Git & GitHub, GeoServer, CrewAIs
- Other: Deep Learning, Machine learning, CNN, LLM, API Development (Flask, REST APIS)

EXPERIENCE

- Createch Software Pvt Ltd — Oct 2024 - Present
  Software Developer (Python & AI)
  - Developed an intelligent chatbot with open-source search and RAG-based chat, enhanced with advanced filters. Integrated audio, text and document translation features, along with map-based search and timeline functionalities.
  - Developed a RAG system with Qdrant for PDF based document querying.
  - Designed frontend UIs for internal apps using Javascript, HTML and CSS.
  - Created CRUD apps for offline systems with seamless data operations using REST APIS.
  - Built a LLM chatbot using Flask & Ollama for secure, on-device inference.
  - Built an LLM based app that converts natural language prompts into SQL queries to interact with a predefined database.
  - Used geoserver for displaying offline based maps.

PERSONAL PROJECTS

- **FashionMNIST Image Classifier(PyTorch)** - https://github.com/Har02sh/fashion-mnist-mlp-pytorch
  Built a fully-connected neural network (MLP) from scratch in PyTorch to classify FashionMNIST clothing images. Implemented data normalization, custom train/validation split, early stopping, best-model checkpointing, and visualization of training metrics. Achieved **~88% test accuracy** with a clean, modular deep-learning pipeline.
- **CIFAR-10 Image Classifier (CNN | PyTorch)**
  Designed and implemented a Convolutional Neural Network in Pytorch for CIFAR-10 image classification with data augmentation, batch normalization, and dropout. Built custom training and evaluation pipelines with learning rate scheduling and modular model design.
- **RAG Chatbot (Ollama + Qdrant + Flask)** - https://github.com/Har02sh/RagChatbot
  Built an offline RAG system with PDF ingestion, embedding, and semantic search using langhchain, Ollama and Qdrant, integrated with a Flask backend for context-aware question answering.
- **Multi-Agent Workflow Builder (React + Flask + CrewAI)**
  Built a drag-and-drop interface for creating multi-agent pipelines (Research, Summarize, Analysis, Custom). Orchestrated sequential agent execution using crewAI with step-by-step reasoning logs.

CERTIFICATIONS

- Python Programming – CEC Certified
- Computer Networks – NPTEL Certified

EDUCATION

- Jagannath Institute of Management Sciences — 2022
  Bachelor of Computer Applications